

Stars and galaxies

GALAXIES ARE HUGE STAR SYSTEMS, MADE UP OF STARS AND LARGE AMOUNTS OF GAS AND DUST.

Our Solar System is just one of billions in our local area, or galaxy. Our galaxy, the Milky Way, is one of hundreds of billions in the Universe.

Astronomical objects

Astronomers estimate that about 6,000 objects can be seen from Earth with the naked eye. Most appear as points of light, but a closer look through a powerful telescope reveals that there is a lot more than just stars and planets out there.



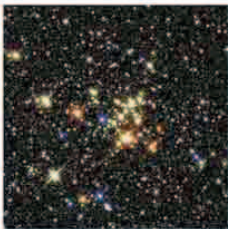
The Sun

Our local star is the source of almost all light and heat reaching Earth. However, it is a very average star, in terms of size and temperature. The next nearest star to Earth is Proxima Centauri, which is 4.2 light-years away.



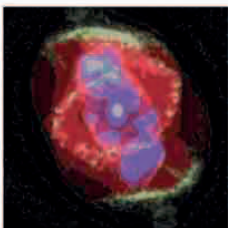
Constellation

Ancient astronomers organized the visible stars into patterns called constellations, most based on images from Greek mythology, such as Ursa Major. Although the stars in a constellation look close together, they are at vastly differing distances from Earth.



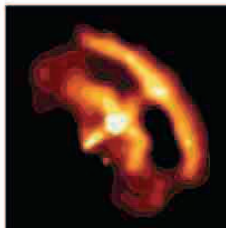
Star cluster

Stars are seldom found alone; most travel either with one or more companions. Pairs are called binary stars. Small groups of stars are called clusters and are made up of stars that formed at the same time, held together in groups by gravity.



Planetary nebula

As some stars grow older they eject their outer layers to become planetary nebulae. The glowing, colored rings of hot gas and dust make stars such as the Cat's Eye Nebula some of the most stunning objects in space. The faint star at the center is known as a white dwarf.



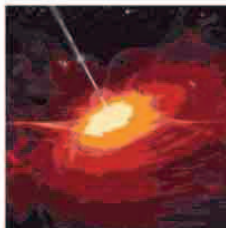
Pulsar

Some dying stars blow apart in a massive explosion called a supernova. The core of the star may collapse to form a small, dense neutron star that emits beams of energy and rotates at amazing speeds. If the beams are detected on Earth, the star is known as a pulsar.



Galaxy

Galaxies are vast star systems that exist in a range of shapes and sizes. The smallest have a few million stars, and the largest, several trillion. Around half of galaxies are spiral-shaped disks, with a central bulge and arms spiraling away from it.



Quasar

Among the most distant of all objects are quasars—young galaxies seething with energy as billions of stars form. Light from distant objects takes many billions of years to reach Earth, so we see quasars as they were when the light left them all that time ago.

REAL WORLD

The Milky Way

Our galaxy is called the Milky Way. It is a spiral galaxy, but from Earth we see it as a pale strip across the night sky. The ancient Greeks called this the *galaktikos*, which means "milky path." The Solar System lies in an outer arm.



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- ◀ 178–179 Gravity
- ◀ 194–195 Electromagnetic waves
- ◀ 230–231 Astronomy
- ◀ 232–233 The Sun